

Chain Rule Worksheet

For each problem, draw a derivative tree and find the derivatives.

1. If $z = x e^{xy}$, $x = t^2$, $y = \frac{1}{t}$, find $\frac{dz}{dt}$

2. If $w = \cos(yx^2)$, $x = t^4 - 2t$, $y = 1 - t^6$, find $\frac{dw}{dt}$

3. If $z = x^2 y^3 + y \cos(x)$, $x = \ln(t^2)$, $y = \sin(4t)$, find $\frac{dz}{dt}$

4. If $f(r, \theta) = e^{2r} \sin(3\theta)$, $r = st - t^2$, $\theta = \sqrt{s^2 + t^2}$, find $\frac{\partial f}{\partial s}$ and $\frac{\partial f}{\partial t}$

5. If $w = \frac{x^2 - z}{y^4}$, $x = t^3 + 7$, $y = \cos(2t)$, $z = 4t$, find $\frac{dw}{dt}$

6. If $u = x^{-2} y^6 - 4x$, $x = t_1^2 t_2$, $y = t_2 - 3t_1$, find $\frac{\partial u}{\partial t_1}$ and $\frac{\partial u}{\partial t_2}$

7. If $w = x \tan^{-1}(yz)$, $x = \sqrt{u}$, $y = e^{-2v}$, $z = v \cos(u)$, find $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$

8. If $w = \sqrt{x^2 + y^2} + \frac{6z}{y}$, $x = \sin(p)$, $y = p + 3t - 4s$, $z = \frac{t^3}{s_2}$, $p = 1 - 2t$, find $\frac{\partial w}{\partial t}$, and $\frac{\partial w}{\partial s}$